Table 1

ESP32 + SD Module — Round 2 Powered Electrical Tests

⚠ WARNING: USB POWER MUST BE connected FOR ALL ROUND 2 TESTS. USE DC VOLTAGE MODE ONLY.

ESP32 and SD module are POWERED. NEVER use resistance (Ω) or current (A / mA) mode on the powered circuit.

Date	03-10-2026			**Multimeter**	DT 830B	
ESP 32	✓			**Meter Range**	20 DCV	
SD Module	✓					

Test #	Test	Test Description	Expected	Actual	Result	Notes
1	SCK Idle State	Whether SCK rests LOW when SPI is idle	0V	0V	✓ Pass	SDSCLK Red / SDGND black SD Inserted
2	SCK Activity	Whether ESP32 actually generates clock pulses	Activity during SPI transmission	1.26V	✓ pass	SDSCLK Red / SDGND black SD Inserted
3	MOSI Idle State	Resting state of the MOSI line	**~0 V** when no data is being transmitted	0V	✓ pass	SDMOSI / SDGND black SD Inserted.
4	MOSI Activity	Whether ESP32 actually sends data on MOSI	Voltage should show activity/ change when data is transmitted	1.57V	✓ pass	SDMOSI / SDGND black SD Inserted
5	CS Idle State	Whether CS rests HIGH when no device is selected	**~3.3 V** (HIGH when SD is not selected)	3.27V	✓ pass	SDCS / SDGND black SD Inserted.
6	CS Selection Behaviour	Whether CS goes LOW when the SD card is selected	CS should transition from ~3.3 V HIGH to ~0 V LOW during an SD transaction	1.66V	✓ pass	SDSS / SDGND black SDI
7	MISO Idle State	Whether the SD card's response line has an appropriate idle state	Should normally remain HIGH/defined when the SD card is not responding	0V	observation ~	Multimeter cannot determine weather it is faulty or not.
8	MISO Response Activity	Whether the SD card changes MISO in response to SPI traffic	MISO should change/ respond when the SD card receives a command	0V	NO response	Potential issue found SDMISO / SDGND black SD SD inserted.
9	SPI Clock Configuration	Verify the ESP32 is actually configured for the intended SPI frequency/mode	Configured SPI frequency should match the test specification (400 kHz)	expected	✓ pass	No probes needed.
10	SPI Mode Verification	Verify the test is using **SPI Mode 0**	SPI Mode 0	expected	✓ pass	No probes needed
11	SPI Transaction Test	Send controlled SPI bytes and verify the ESP32 transmits them	ESP32 successfully transmits controlled SPI bytes	expected	✓ pass	No probes needed.

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ESP32 + SD Module — Round 2 Powered Electrical Tests						
12	SD Command Response	Send CMD0/CMD8 and document the actual returned bytes	CMD0 should return R1 = 0x01 during SD SPI initialization	expected	✓ pass	Significant meaning it can communicate for SD command. no probes needed.
13	Repeated SPI Communication	Repeat the communication test to establish whether the responses are consistent	Responses should be consistent across repeated tests	5/5 attempts	✓ pass	Response is consident across all.
14	Round 3 Summary	Consolidate all SPI evidence before moving to the next round	All available SPI evidence documented before proceeding			

→ Conclusion :

communication between esp32 and SD card module, is demonstrated and it is repetable

Important findings:

S8 → Multimeter saw no MISO activity

S11 → generic SPI RX bytes were all 0x00

S12 → actual SD CMD0 returned 0x00

S13 → CMD0 returned 0x01 5/5 Times.

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SPI Simultenous sending and receiving data

Rx = Receive } data receive from SD card

Tx = Sending } data send to SD card.

0x000 is hex = 000000000

CMD0 command 0 SD card protocol, the command used to Initialise communication with SD card in SPI mode.

R1 = 0x01 meaning it is in Idle state.